

University of Delaware – Computer Science

CISC 442/642

Project Details

The idea of this project is to work together using tools discussed in this class to create a cool computer vision project. You can use this as a learning opportunity to implement algorithms you may be interested in but did not get a chance to create yet, or build an application. You are given the freedom to choose the project and how to handle it, but must follow the guidelines listed below.

Team up with up to 4 people.

Have Fun and Learning Something

The most important part of this project is to have fun with it. Computer vision is super futuristic and cool, so be creative and have fun with it. The whole point of this is to learn something new, whether it be an algorithm you did not quite fully understand, or combining CV tools to create something unique.

Project Proposal -- Due by April 24th on Canvas

Write a ½ to 1-page proposal detailing the who, what, and how of the project you intend to work on. This means list out the group members (1-4 people), the idea you want to implement, and how you think you can implement it. What data will you use, what tools will you use, what algorithms will you use? This is where you do some background research and thinking to make sure you don't give yourself too much or too little work, and you write down and semi-formalize your plans.

Project Presentation – On May 5th

May 5th, 7th, 12th and 14th we will dedicate to project presentations. You and your team will give a 10-15 minute presentation on the project you implemented, summary of the problem, how you solved it, algorithms implemented or employed, technical hardships you encountered, and a demo/video of results.

Code + Documentation Uploaded to Canvas – On May 14th

Please upload code with an example image/video/data to run it on. This should have instructions on how to run it and should be easy to get working to test it. **PLEASE CITE**

RESOURCES/TOOLS YOU USED. Using available tools and code is fine and even encouraged, but do not claim others work as your own. I want to know how much is yours vs how much is from online resources.

Grading

Grading is somewhat subjective, but I will consider the following to decide a final grade:

Difficulty and Size of Project: The project should be an appropriate size for how many people in your group. E.g. If you have 4 people, but you can solve your problem by simply downloading and running code available online, that is not appropriate and will result in a low score. You may implement known algorithms from scratch, but if you download code and just run it for your final answer, that is low effort.

Quality of Code/Documentation: I expect working code that I could run on new images/videos and get a result. It should be well-documented and easy to run and test. I don't expect your code to work on all possible images, but the more robust the better.

Quality of Presentation and Proposal: How thorough was your planning and adjustments to problems you encountered? How clear is it what you intend to do and if you followed through? Is it easy to understand the problem you are tackling and your solution?

Example Ideas

Although you are free to decide what problem you would like to work on, below are a few examples.

- When I took this class, we used the ARtoolkit and OpenGL to create a few Augmented Reality games. The system would recognize markers we created and draw 3D models on top of the markers. We created an AR jigsaw puzzle that would recognize when the puzzle was complete and display an AR “You win” message. The other game was “Face Wars” and using 3D models of our faces created with a shape from shading iPhone app, lasers would shoot out of our AR mouths into the scene. Using line-model intersection we detected if the character was hit by the laser and would change the model to be a dead” version of our face.

Android and iPhone both have marker-less AR tool kits, I downloaded arCore the other day and have a “hello world” app working natively on my phone.

- One idea is to create a stereo camera setup and get it working live to 3D reconstruct the class as a demo. This would include physically creating a stereo camera setup on a baseline, calibrating, rectifying, stereo matching, and 3D reconstruction.
- Deep learning is the current state of the art in many areas of CV. Get familiar with deep learning by creating a few networks with tensorflow + Keras and learning some CV task such as classification or detection. Warning: this requires a good GPU and a lot of data for training, only try if you think you have the hardware to get this working (this will involve some research). Alternatively, use already trained networks to create some larger CV system.
- Go out to main street and record videos and count people walking by. Detection w/ bounding boxes + tracking. Write up how accurate CV is vs ground truth.
- Feel free to double dip if you are a grad student and working on a research project.
- Make a cell phone app that implements some CV algorithm and sell it to make \$1,000,000.